Chapter 2 in [1]

Exercise (2.1-7)**.** Let a random experiment be the casting of a pair of unbiased six-sided dice, and let X equal the smaller of the outcomes if they are different and the common value if they are equal.

1. With reasonable assumptions, find the p.m.f. of X.
2. Let Y equal the range of the two outcomes (i.e. the absolute value of the difference of the largest and the smallest outcomes). Determine the p.m.f. $g(y)$ of Y for $y = 0, 1, 2, 3, 4, 5$.

Solution. 題目表示出 X 是六面骰骰兩次中最小點數的隨機變數。因爲合計有 *36* 種可能狀況，其中可以知道數字爲 *1* 的情況有 *11* 種，以此類推。更具體一點爲，考慮 $X = x$，那麼我們可以將

$$P(X \geq x) - P(X \geq x + 1) = P(X = x)$$

用來計算，因此

$$(7 - x)^2 - (6 - x)^2 = 13 - 2x$$

所以 X 的 *p.m.f.* 爲

$$\frac{13 - 2x}{36}$$

而 Y 的 *p.m.f.* 則可以直接計數 *(*暫時沒想到其他辦法*)* 得到

$$P(Y = y) = \begin{cases} \frac{1}{6}, & y = 0, \\ \frac{6-y}{18}, & y = 1, 2, 3, 4, 5. \end{cases}$$

□

Exercise (2.1-11)**.** In a lot (collection) of 100 light bulbs, there are 5 bad bulbs. An inspector inspects 10 bulbs selected at random. Find the probability of finding at least one defective bulb. Hint: First compute the probability of finding no defectives in the sample.

Solution. 抽 10 個至少壞 1 個的機率

$$P = 1 - \text{全部都是好的機率} = 1 - P'$$

全部是好的機率就很好算了。

$$P' = \frac{95}{100} \times \frac{94}{99} \cdots \times \frac{86}{91}$$

藉由計算機，

$$P = 0.4163$$

□

Exercise (2.1-13)**.** Let X be the no. of accidents per week in a factory. Let the p.m.f. of X be

$$f(x) = \frac{1}{(x + 1)(x + 2)}, \quad x = 0, 1, 2, \cdots .$$

Find the conditional probability of $X \geq 4$, given that $X \geq 1$. Hint:Write $f(x) = 1/(x + 1) - 1/(x + 2)$.

Solution.

$$P(X \geq 4 | X \geq 1) = \frac{P(X \geq 4 \cap X \geq 1)}{P(X \geq 1)} = \frac{P(X \geq 4)}{P(X \geq 1)}$$

那麼先算 $X \geq 4$，

$$P(X \geq 4) = \sum_{4}^{\infty} \frac{1}{n + 1} - \frac{1}{n + 2} = \frac{1}{5}$$

以及

$$P(X \geq 1) = \sum_{1}^{\infty} \frac{1}{n + 1} - \frac{1}{n + 2} = \frac{1}{5} = \frac{1}{2}$$

所以答案爲 $\frac{2}{5}$。 □

References

[1] R. V. Hogg and E. A. Tanis, *Probability and Statistical Inference*, Pearson, 2010.